

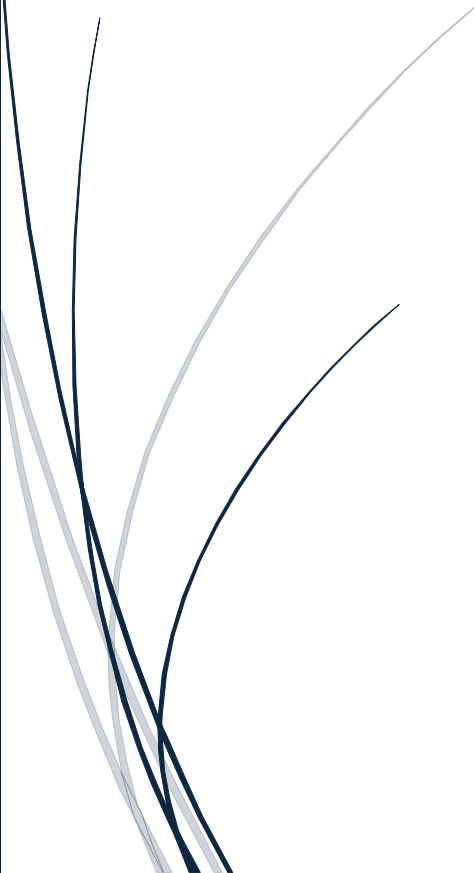


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Project Report

SaaS Data Pipeline (Revenue & Customer Retention)

Tech Stack: PostgreSQL (Database), DBeaver (ETL/SQL), Power BI (Visualization)



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1. Project Overview & Objective

This project was designed to simulate a real-world SaaS (Software as a Service) environment. The goal was to build a robust data pipeline that transforms raw, "dirty" data into high-level business intelligence. By generating a synthetic dataset of 100 users across Germany, France, the UK, and the USA, I created a scenario to test data cleaning, relational modeling, and financial metric calculation (MRR, ARPU, and Churn).

2. Data Simulation & The "Dirty Data" Challenge

To mirror real-life database issues, I intentionally injected several data quality problems into the initial SQL script (`01_Data_Generation`):

- **Duplicate Transactions:** Injected 15 "double-billing" errors where the same subscription was charged twice at the same second.
- **Inconsistent Naming:** Used a mix of country codes and full names (e.g., 'DE', 'Deutschland', 'Germany').
- **Status Casing:** Subscription statuses were a mess of 'active', 'Active', and 'Active_User'.
- **Payment Failures:** Included 'Failed' and 'Pending' statuses that should not be counted as revenue.

3. The SQL Transformation Layer (ETL)

In the cleaning script (`02_cleaning_and_views`), I built a series of **PostgreSQL Views** to act as an automated filter.

3.1 Deduplication with Window Functions

The most critical fix was removing the 15 duplicate transactions. I used a **Common Table Expression (CTE)** and the `ROW_NUMBER()` window function:

- **Logic:** I partitioned the data by `sub_id`, `payment_date`, and `amount_eur`.
- **Result:** This assigned a '1' to the original and a '2' to the ghost duplicate, allowing me to filter for only `row_num = 1`.

3.2 Standardization & Normalization

- **Country Mapping:** Used `CASE WHEN` logic to map all variations ('DE', 'DEU') into a single 'Germany' label.
- **Financial Integrity:** Standardized all statuses to lowercase and filtered out unsuccessful payments to ensure the MRR was 100% accurate.

4. Problem Solving: The "Static View" Hurdle

During the transition to Power BI, I encountered a significant challenge. My initial summary views (`vw_mrr_report` and `vw_churn_report`) were pre-aggregated at a global level.

The Problem: When I added a "Country Slicer" in Power BI, the charts wouldn't move because the summary views didn't contain a `country` column—they were "disconnected" from the user data.

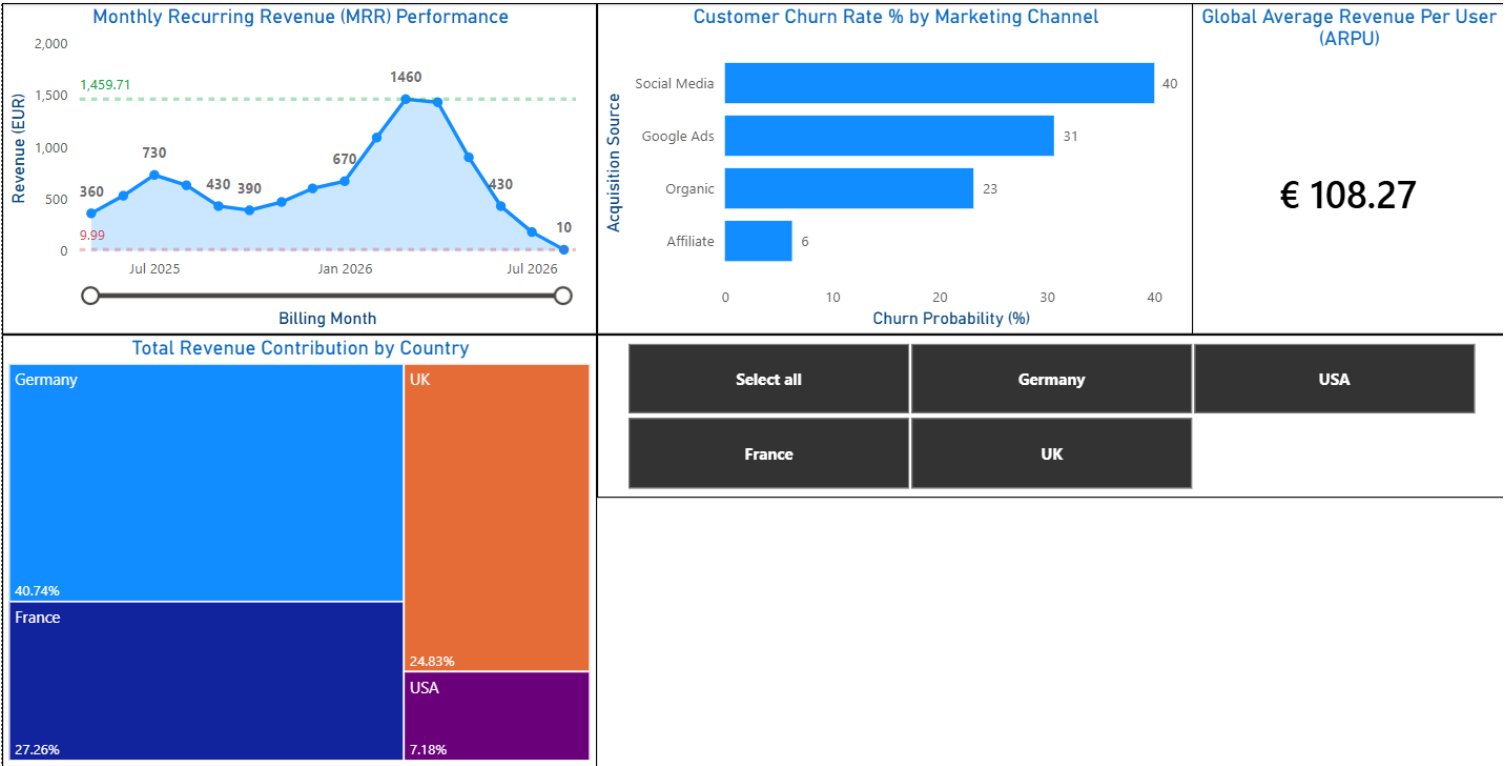
The Solution: I refactored the SQL code to "inject" the geographic dimension back into the summary views using `JOIN` statements. By dropping the old views and recreating them with `u.country_clean`, I enabled a **Many-to-Many relational model** in Power BI that allowed every chart to respond to the slicer perfectly.

5. Final Audit & Performance Metrics

After running a final SQL audit, the business health is summarized as follows:

Metric Scope	Total Revenue	Average ARPU	Peak Monthly MRR	Avg Churn Rate
Global	€9,907.59	€97.55	€609.89	15.76%
Germany	€4,159.16	€126.04	-	-
UK	€2,839.41	€118.31	-	-
USA	€1,309.57	€81.85	-	-
France	€1,599.45	€63.98	-	-

6. Power BI Dashboard Analysis



Key Insights:

- Revenue Growth:** The MRR Performance chart shows a peak of **€1,460** (unfiltered total), highlighting strong growth periods.
- Churn Risk:** **Social Media** (40%) and **Google Ads** (31%) show the highest churn probability. This indicates that while these channels bring in users, they are not as "sticky" as **Affiliate** leads (6%).
- Country Performance:** **Germany** is the clear revenue driver, but the **UK** holds a very high ARPU, making it a highly profitable target for expansion.

4. **Interactivity:** The dashboard includes a **Country Button Slicer** and a **Billing Month Slider**, allowing management to zoom into specific markets or timeframes instantly.

7. Conclusion

This project successfully transformed raw, error-prone data into a professional decision-making tool. It demonstrates a mastery of **Relational Database Design**, **Complex SQL Aggregations**, and **Interactive Visual Storytelling** in Power BI.